

Getting Started with R: Setup and Loading your Data

Lesson 1

Leonard N., Dam K., Noutin M
18th September 2025

In this course (5 hours 15 min) you will learn:

- How to use R studio to write R code
- R syntax, data types, operators and functions
- Data wrangling*
- Visualizations

Introduction to R language

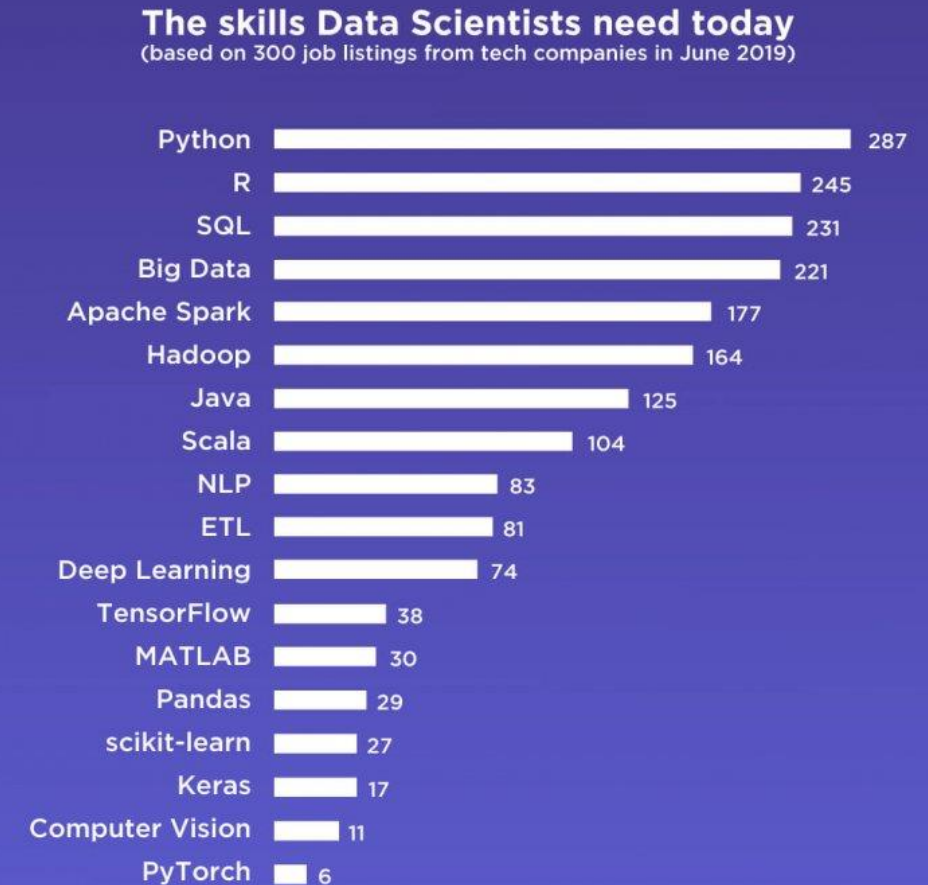
- R is a **language** and **environment** written and released by Ross Ihaka and Robert Gentleman in the 90s for statistical computing and graphics-facilitated by packages (21,203)
- **R**: effective data handling, a suite of operators, collection of packages for data analysis, graphical facilities for data visualization
- **R**: open source software. Not only is the software free but packages are also free.

Introduction to R Studio

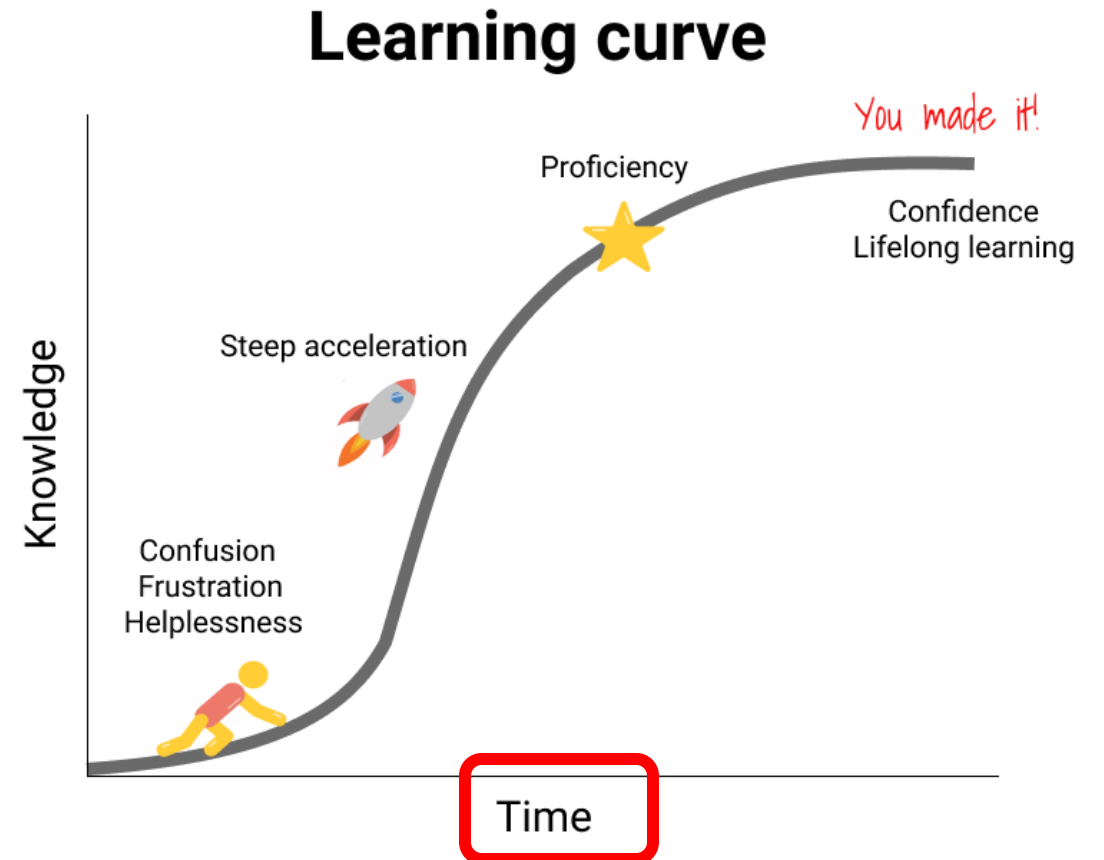
- R Studio is also a free and open source integrated development environment (IDE) for R programming language for statistical computing and graphics.
- R and R Studio are two different versions of the same thing.
- R is a programming language and R Studio is a IDE that has a more GUI to make analytics easy.
- We can use R without R Studio but we cant use R Studio without R.

R: Among many other Programming Languages

Programming Language	Data Science Tasks Each Performs
 python	• Conducts data mining • Carries out ML algorithms
	• Performs data visualization • Conducts Data Analytics
	• Eases the performance on high datasets • Sculpts data in any given form
	• Solves mathematical complications at high speed • Performs Data Analytics
	• Wise option for IoT and Big Data • Secure enough to work with sensitive data
	• Manages large databases • Compliant toward Data Science workflow
	• Performs profound mathematical operations • Highly specialized in working with Big Data
JavaScript	• Sets up data visualizations perfectly • Good fit for projects based on web and Big Data technologies
	• Manipulates and manages data • Administers data analysis through statistical models
	• Used in Big Data in collaboration with Java • Computes large datasets quickly



Patience | Time | Practice | Learning from Mistakes



Common applications of R

- I have two datasets, and I want to merge them
- I have a very large file (Excel can't open) how do I select a few columns to work with?
- Data transformations
- Data cleaning
- 16S analysis: *DADA2*
- RNA-transcriptomics analysis: *DESeq2*
- Statistical analysis
- Visualizations: how to I generate publication ready plots?
- Spatial analysis: creating maps
- Dashboards and interactive interfaces
- **Other examples?**

Where do I start

Step 1: Downloading & Install R (<https://cran.r-project.org/>)

Step 2: Download and install R studio (user interface) (<https://posit.co/download/rstudio-desktop/>)

The Comprehensive R Archive Network

Download and Install R

Precompiled binary distributions of the base system and contributed packages, **Windows and Mac** users most likely want one of these versions of R:

- [Download R for Linux](#) ([Debian](#), [Fedora/Redhat](#), [Ubuntu](#))
- [Download R for macOS](#)
- [Download R for Windows](#)



OS	Download	Size
Windows 10/11	RSTUDIO-2024.04.2-764.EXE 	262.79 MB
macOS 12+	RSTUDIO-2024.04.2-764.DMG 	664.40 MB
Ubuntu 20/Debian 11	RSTUDIO-2024.04.2-764-AMD64.DEB 	194.73 MB

Now we open R studio

Lesson01.R* × data_01_kenyaGC ×

Source on Save 🔍 🛠️ 📄

Run ↗️ ⬆️ ⬇️ Source ▾

```
1 #=====#
2 # Installing packages
3 #=====#
4 library(data.table)
5 library(dplyr)
6 library(stringi)
7 library(stringr)
8 library(janitor)
9 library(tidyverse)
10 library(ggplot2)
11 library(readxl)
12 library(openxlsx)
13 library(ggpubr)
14
15
16 |
17
18
19
20
21
22
23
24
25
26
27
28
29
```

16:1 (Top Level) R Script ▾

Editor

Console Terminal × Background Jobs ×

R 4.3.2 · ~/Documents/Tutorials/WellcomeConnectingScience/2024-Course/ ↗️

Copyright (c) 2023 The R Foundation for Statistical Computing
Platform: aarch64-apple-darwin20 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Workspace loaded from ~/Documents/Tutorials/WellcomeConnectingScience/2024-Course/.RData]

> |

Console: Script output
panel(errors & warnings)

Environment History Connections Tutorial

Import Dataset ▾ 120 MiB 🛠️

R ▾ Global Environment ▾ 🔍

Data

data_01_afriMenin	716 obs. of 22 variables	📅
data_01_kenyaGC	103 obs. of 10 variables	📅

Data, Objects, History

Files Plots Packages Help Viewer Presentation

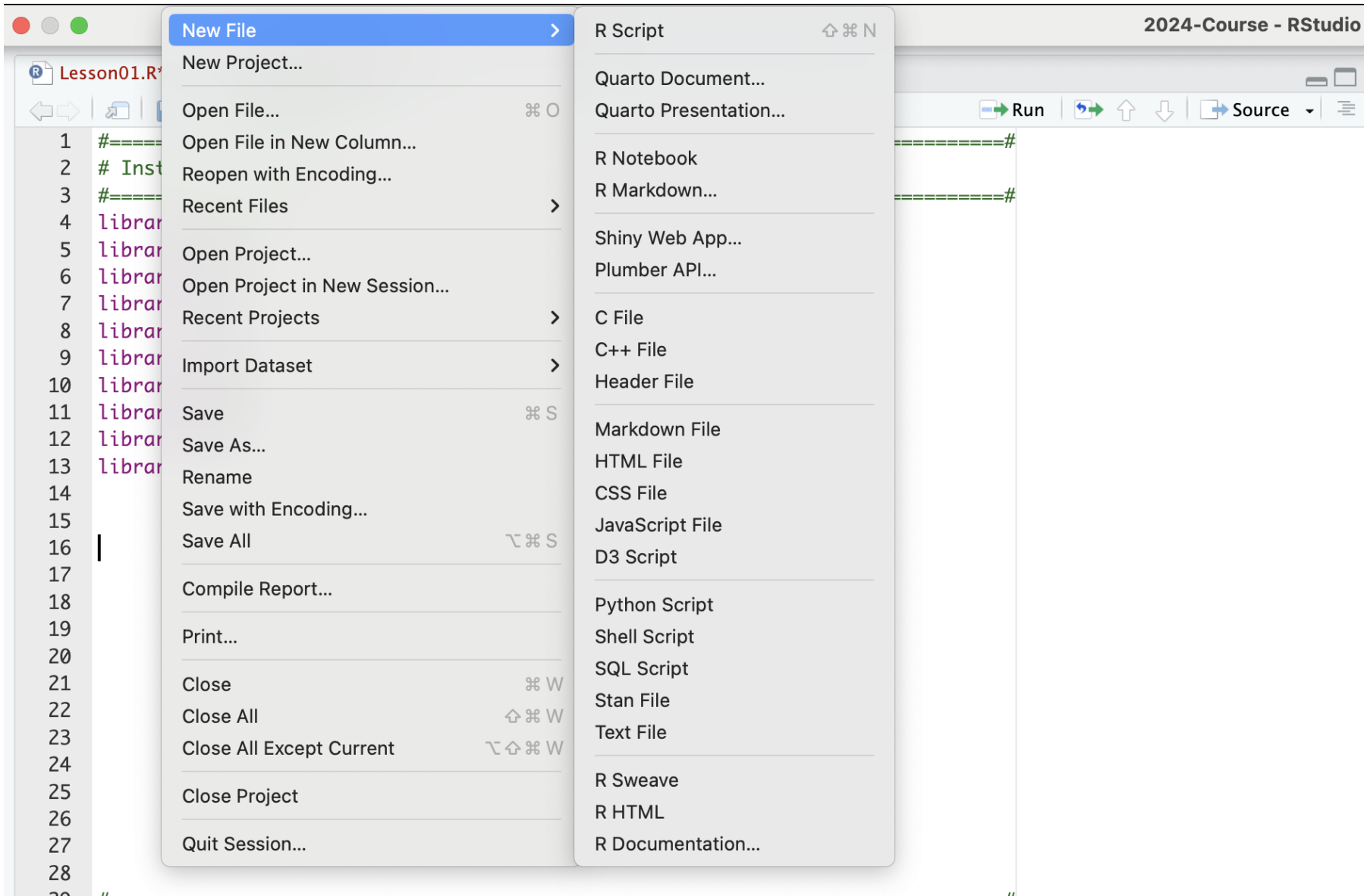
New Folder + New Blank File ▾ Delete 🗑️ Rename 🔄 More ⚙️

🏠 Home > 📁 Documents > 📁 Tutorials > 📁 WellcomeConnectingScience > 📁 2024-Course > 📁 datasets 🔍 ⋮

	▲ Name	Size	Modified
📁	..		
☐	📄 African_meningococci.xlsx	79.2 KB	Jul 11, 2024, 4:39 PM
☐	📄 Kenyan_GC_datasetv2.xlsx	14.4 KB	Jul 11, 2024, 4:39 PM
☐	📄 SILVA_132_SSURef_Nr99_tax_silva.fasta	1 GB	Jul 22, 2024, 10:52 AM

Files, Plots, Package, help

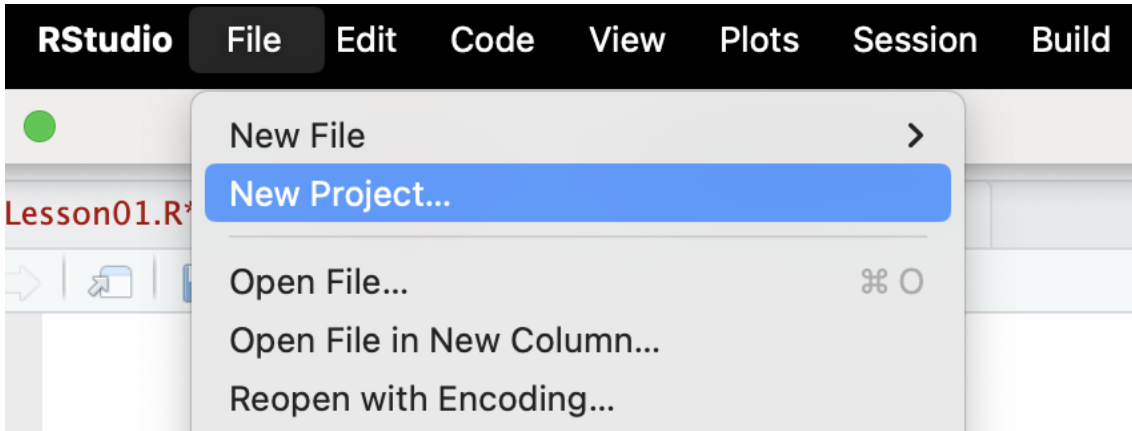
R studio can be used with other Languages



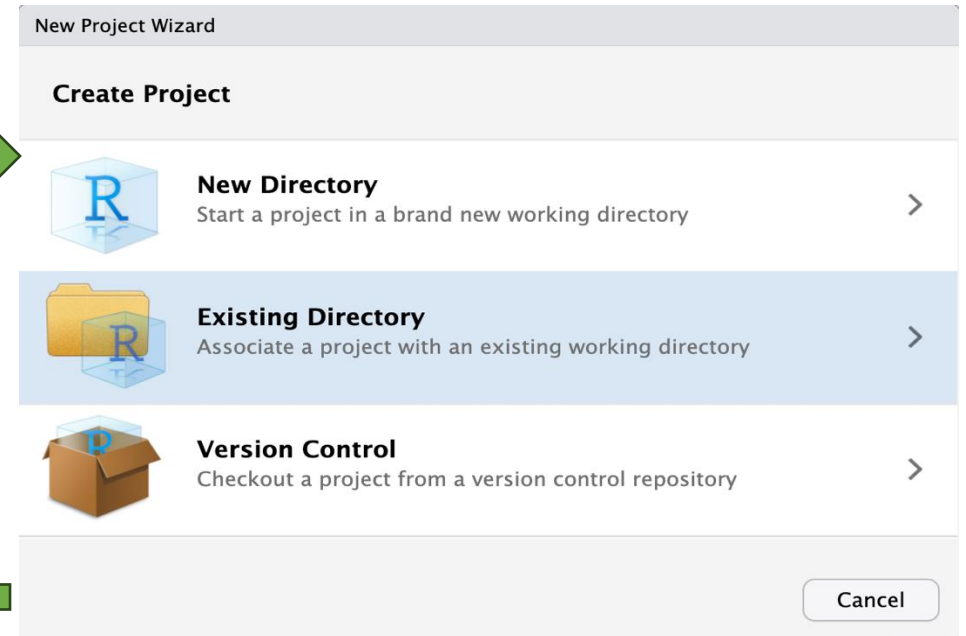
R studio: User interface, Customizable interface, Code autocompletion, write code in other languages

Creating an R project

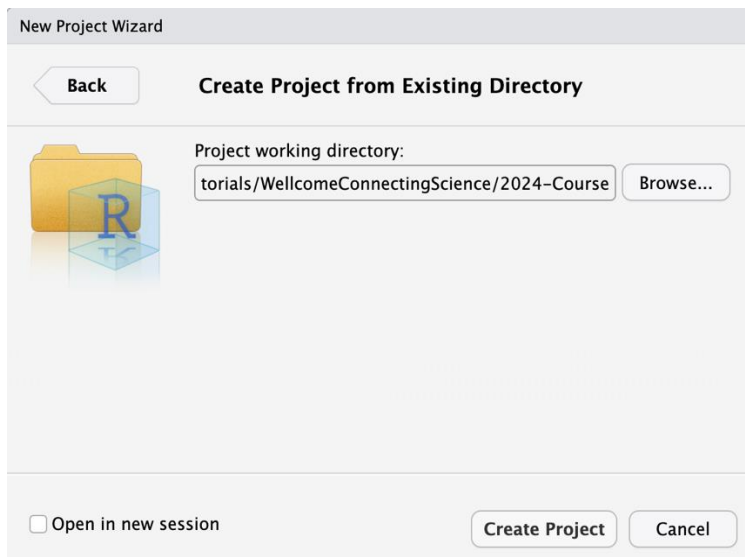
Step 1



Step 2

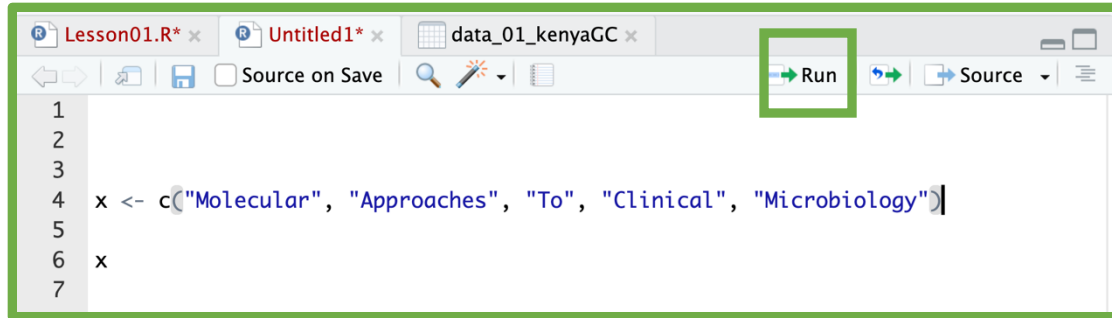


Step 3



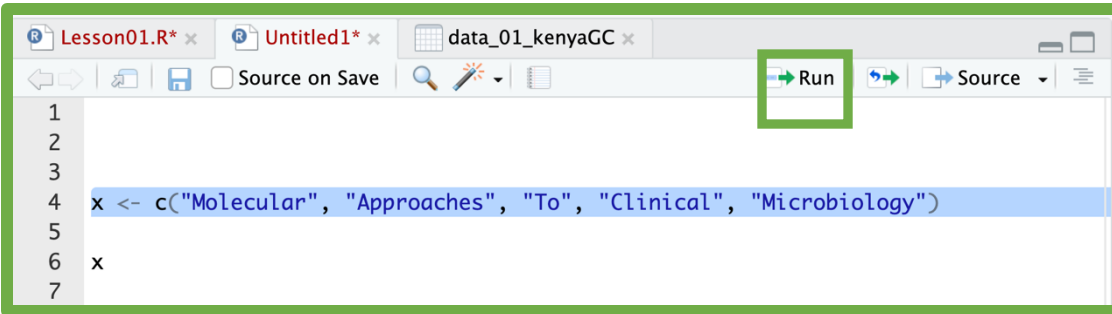
Running your Code

1



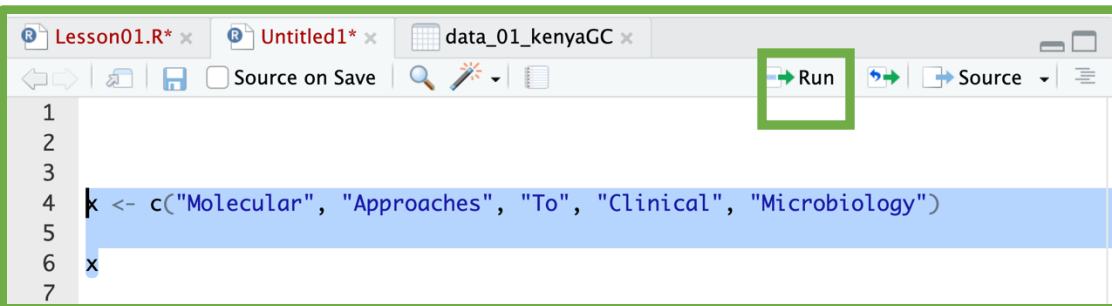
```
1  
2  
3  
4 x <- c("Molecular", "Approaches", "To", "Clinical", "Microbiology")  
5  
6 x  
7
```

2



```
1  
2  
3  
4 x <- c("Molecular", "Approaches", "To", "Clinical", "Microbiology")  
5  
6 x  
7
```

3

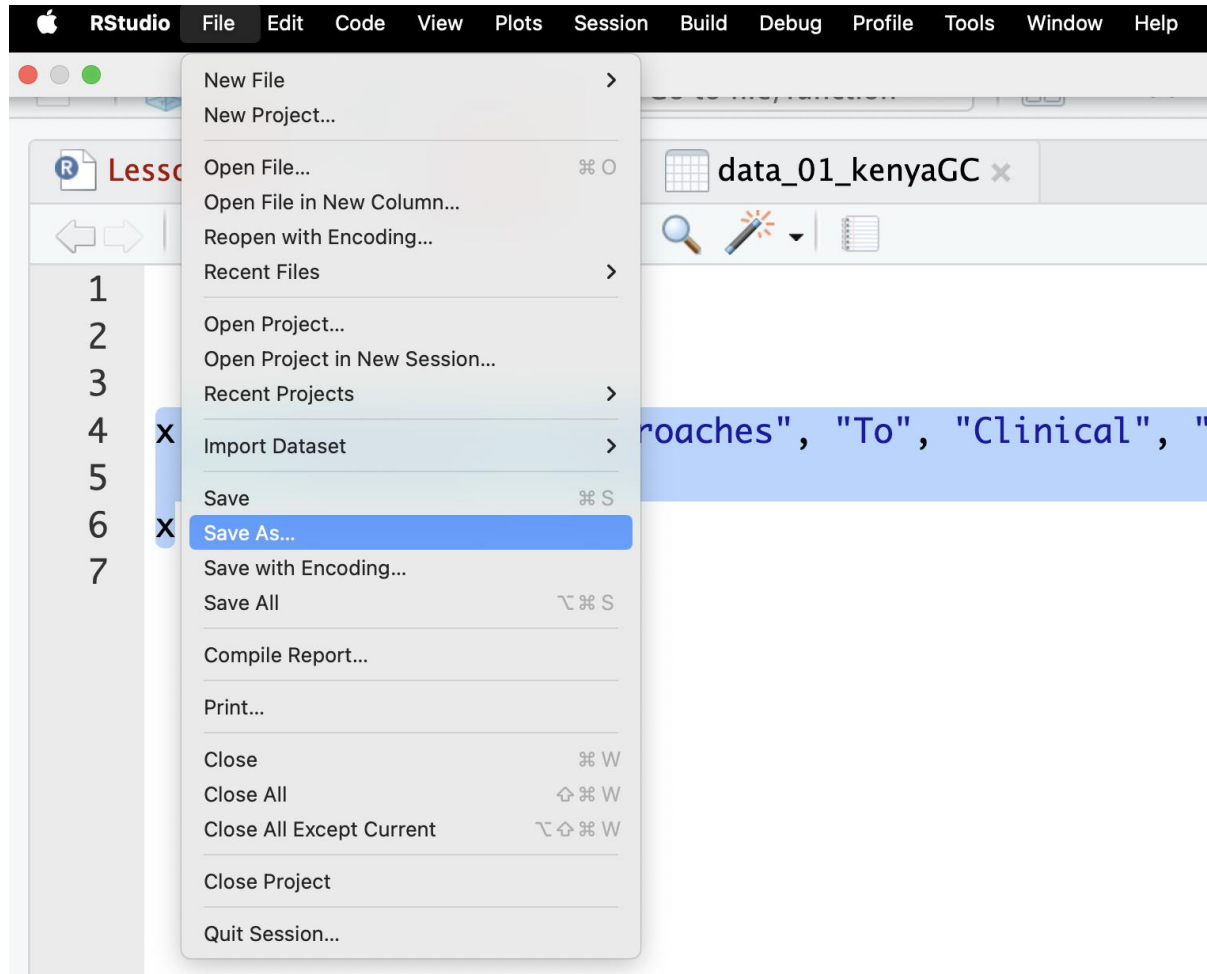


```
1  
2  
3  
4 x <- c("Molecular", "Approaches", "To", "Clinical", "Microbiology")  
5  
6 x  
7
```

Or use:

Ctrl + Enter
instead of
pressing the
"Run" button

Saving your code



Save As:

Untitled.R

Tags:

Basic but Important Commands

1. # - not a command, but used for commenting you code

```
#=====#  
# Course Title: Molecular and Genomic Approaches to Clinical Microbiology in Africa  
# Date: 13-19 September 2025  
# Location: MRC Unit, The Gambia  
#=====#
```

Basic but Important Commands

2. Installing packages: **install.packages("name_of_package")**
and loading packages: **library(name_of_package)**

```
# Installing the required package  
install.packages("ggplot2")
```

All the packages we require for this training have been installed.

```
# Loading the required packages  
library(ggplot2)
```

But note that you need to load the packages

Use the function **search()** to know which packages have been loaded

Basic but Important Commands

1. **#** - not a command, but used for commenting you code
2. Installing and loading packages:
install.packages("name_of_package") then
library(name_of_package)
3. **setwd()** & **getwd()**

Assigning variables (working on the console)

```
> x <- 10
```

```
> x
```

```
[1] 10
```

Method 1: Most preferred

```
> x=10
```

```
> x
```

```
[1] 10
```

Method 2

```
> assign("x", 10)
```

```
> x
```

```
[1] 10
```

Method 3

Assigning multiple values to a variable (working on the console)

```
> x <- 10, 20, 30, 40|
```

```
> x <- 10, 20, 30, 40
```

```
Error: unexpected ',' in "x <- 10,"
```

```
> x <- c(10,20,30,40)
```

```
> x
```

```
[1] 10 20 30 40
```



Creating a data frame (working on the text editor)

month	male	female
Jan	20	40
Feb	30	30
March	40	20

```
male <- c(20, 30, 40)
```

```
female <- c(40, 30, 20)
```

```
month <- c("Jan", "Feb", "March")
```

```
df <- data.frame(male, female, month)
```



function

Simple Statistics (working on the text editor)

mean|

mean	{base}	mean(x, ...)
mean.Date	{base}	Generic function for the (trimmed) arithmetic mean.
mean.default	{base}	Press F1 for additional help
mean.difftime	{base}	
mean.POSIXct	{base}	
mean.POSIXlt	{base}	
mean_ci	{ggpubr}	
mean_cl_boot	{ggplot2}	

mean(df\$male)

Task: Try and get the **median** and **sum**. Use the function 'summary' to summarise the entire dataset

Bioinformatics Eyesight Test

 `install.package(data.table)`

 `install.packages(data.table)`

 `install.packages("data.table")`

 `library(data.table)`

 `library("data.table")`

 `library(data.table)`

 `require(data.table)`

```
x <- 20
y <- 30
z <- 40
```

 `female -> c("x", "y", "z")`

 `female -> c(x, y, z)`

 `female <- c("x", "y", "z")`

 `female <- c(x, y, z)`

 `month <- c("Jan" "Feb" "March")`

 `month <- c(Jan Feb March)`

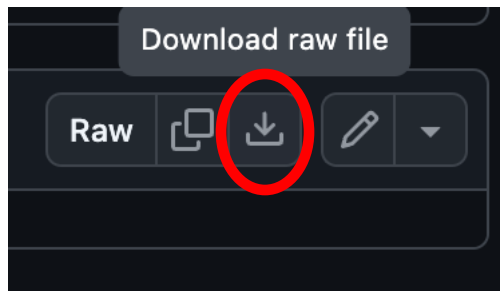
 `month <- c("Jan", "Feb", "March")`

 `female <- c(40, 30, 20)`

Download the Datasets (while inside you VM)

- Github page- <https://shorturl.at/CD5mm>
- Or Access within the virtual machine

Name	Last commit message	Last commit date
..		
 African_meningococci.xlsx	R practical datasets	3 weeks ago
 Kenyan_GC_datasetv2.xlsx	R practical datasets	3 weeks ago



Load datasets

```
read.xlsx(path = "PATH_TO_FILE")
```

```
read.xlsx(xlsxFile = "datasets/African_meningococci.xlsx")
```

```
data_01_Africa <- read.xlsx(xlsxFile = "datasets/African_meningococci.xlsx")
```


Confirm the Dataset is within R

Environment		History		Connections		Tutorial	
 		 Import Dataset ▾		 321 MiB ▾			
R ▾		 Global Environment ▾					
Data				Rows		Columns	
 data_01_Africa				716 obs. of 22 variables			

Environment		History		Connections		Tutorial	
		 Import Dataset ▾	 321 MiB ▾				
R ▾	 Global Environment ▾						
Data							
	data_01_Africa			716 obs. of 22 variables			